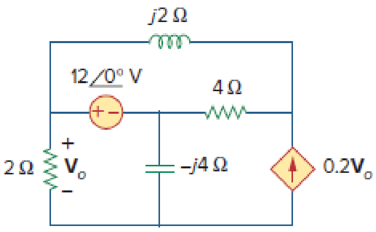
Chapter 10, Problem 19P

Problem

Obtain **V**o in Fig. 10.68 using nodal analysis.

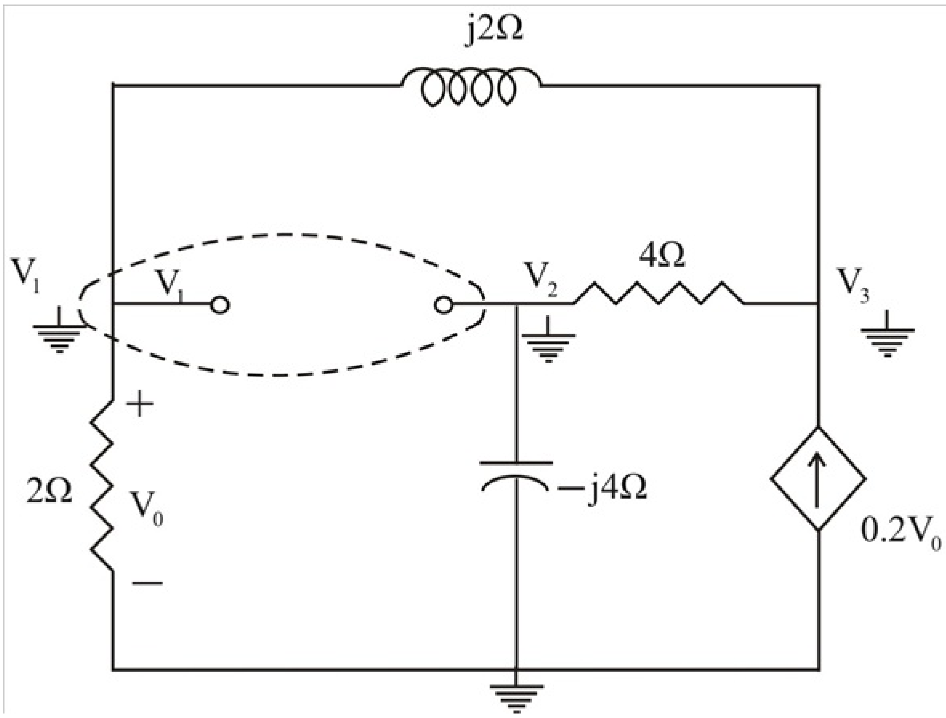
Figure 10.68:



Step-by-step solution

Step 1 of 3

We have a super node as shown in the circuit below.



Step 2 of 3

Notice that $V_0 = V_1$

At the super node,

$$\frac{V_3-V_2}{4} = \frac{V_2}{-j4} + \frac{V_1}{2} + \frac{V_1-V_3}{j2}$$

$$0 = (2-j2)V_1 + (1+j)V_2 + (-1+j2)V_3 \quad \rightarrow (1)$$

At node 3,

$$0.2V_1 + \frac{V_1-V_3}{j2} = \frac{V_3-V_2}{4}$$

$$(0.8-j2)V_1 + V_2 + (-1+j2)V_3 = 0 \quad \rightarrow (2)$$

Subtracting (2) from (1),

$$0 = 1.2V_1 + jV_2 \quad \rightarrow (3)$$

Step 3 of 3

But at the super node,

$$V_1 = 12\angle 0^\circ + V_2$$

Or, $V_2 = V_1 - 12 \quad \rightarrow (4)$

Substituting (4) into (3),

$$0 = 1.2V_1 + j(V_1-12)$$

$$V_1 = \frac{j12}{1.2+j} = V_0$$

$$V_0 = \frac{12\angle 90^\circ}{1.562\angle 39.81^\circ}$$

$$V_0 = 7.682\angle 50.19\ V \quad .$$